

Hiruy Kassa

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EDUCATION

University of St. Thomas

Bachelor of Science in Computer Science

Dec. 2027

3.65/4.0 — Dean's List

- **Relevant Coursework:** Data Structures, Algorithms, Software Design, Computer Architecture, Information Security

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Java, HTML/CSS

Frameworks & Libraries: React, Flask, Node.js, Pandas, BeautifulSoup, pytest

AI / ML: Retrieval-Augmented Generation (RAG), vector embeddings, hybrid search, LLM evaluation

Cloud & DevOps: AWS (EC2, S3, IAM, Lambda, API Gateway, Bedrock, CloudWatch), Docker, GitHub Actions, Linux, Nginx

EXPERIENCE

IT Support & Technician

Jan. 2025 – Present

University of St. Thomas

Minneapolis & St. Paul, MN

- Maintained a **90%+ same-day resolution rate** across **50+ tickets per week** for **3,000+ users** by diagnosing hardware, software, and account issues across Windows and Mac.
- Supported **9,000+ users** across Windows, Mac, and Linux by resolving **15–20 incidents per week** and tracking all requests in TeamDynamix.
- Scaled device rollout capacity to **30+ new workstations per week** during peak summer demand by performing OS imaging, software configuration, and Active Directory domain joins across a multi-campus environment.
- Improved first-contact resolution rates by documenting recurring fixes in the internal knowledge base and escalating complex network and account issues to senior staff.

AI Dark Patterns & Digital Well-Being Research

Jul. 2026 – Present

University of St. Thomas

St. Paul, MN

- Building a reproducible Python (**Pandas**) pipeline to clean and analyze summer-2026 survey data on AI-driven engagement, **documented for replication** by future students.
- Defining the study's engagement metrics and research questions by reviewing HCI literature on dark patterns and digital addiction.

Solutions Architect Intern

Mar. 2026 – Jun. 2026

FZ Contracting

Minneapolis, MN

- Automated **5 manual business processes** by designing a **6-stage agentic AI workflow**, documented in a **22-requirement PRD** with **4 architecture diagrams**.
- Enabled reliable cross-system communication by mapping third-party API integrations (Google Calendar, Google Maps, Twilio SMS) and defining the data flow and error-handling strategy for each stage of the system.

PROJECTS

Ask the Early Church | React, Flask, Python, SQLite/FTS5, Voyage AI, Docker, AWS

Mar. 2026 – Jul. 2026

- Serving **3,000+ monthly users** searching **52,000+ passages** (247 authors, 2,858 works) by a **hybrid search engine** that fuses **vector embeddings**, FTS5 BM25 keyword search, and title matching via **reciprocal rank fusion**.
- Eliminated user-facing API failures by engineering **graceful degradation** across the **LLM stack** (Gemini → Groq → local; vector → keyword fallback), including a monthly budget guard that falls back to keyword-only search at a spend ceiling.
- Cut memory footprint by **~3×** via a custom **float16 embedding loader** (~108 MB) holding the full corpus in RAM, while overlapping Voyage embedding with Gemini parsing in a worker thread and serving repeat queries from 30-day caches.
- Hardened production reliability by deploying on **AWS EC2** via docker-compose (Nginx, gunicorn/Flask, Redis) with SQLite on an EBS volume hydrated from S3, **GitHub Actions CI/CD**, CloudWatch alarms, rate limiting, CSP/CORS, and parameterized SQL.

Lucid: Grounded AI Dark Patterns Assistant | Python, AWS (Lambda, Bedrock, S3), SAM

Jul. 2026 – Aug. 2026

- Developing a **serverless RAG** service (Lambda, API Gateway, **Bedrock embeddings**, **OpenSearch**) that answers AI-dark-patterns questions grounded in ~50 HCI papers with inline citations, currently reaching **~90% faithfulness** and **~85% recall@5** on a 50-question benchmark.
- Iterating on hallucination reduction, currently down from **~18% to ~6%**, by adding chunk reranking and a strict grounding prompt (**LLM-as-judge** eval harness), while holding p95 latency under 2.5s and cost under \$0.01/query; infrastructure codified as **AWS SAM** with GitHub Actions CI/CD.

St. George Church Website | WordPress, PHP

Jun. 2026 – Present

- Kept stgeorgemn.org at **100% uptime** for the parish's full online congregation by partnering with a senior software engineer on ongoing WordPress/PHP development and maintenance.
- Cut parish staff content-update time by an estimated **50%** by building custom WordPress themes and plugins for the events calendar and media galleries, enabling direct self-service edits.
- Reduced security vulnerabilities and compatibility bugs by performing **regular PHP/WordPress core updates** and plugin patching across releases.